Aqib Habib

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EDUCATION

Technical University of Munich, *M.Sc Aerospace Engineering* — Jun 2024 – Dec 2026
Note: 1,8/1,0 — Munich, Germany

National University of Sciences and Technology, — Sep 2023 – May 2024
M.Sc Mechatronics, Robotics, and Automation Engineering — Islamabad, Pakistan
Note: 1,2/1,0

National University of Sciences and Technology, *BE Mechanical Engineering* — Sep 2018 – Jun 2022
Main Note: 1,3/1,0 — Islamabad, Pakistan
Minor in Computer Science with a note of 1,1/1,0

PROFESSIONAL EXPERIENCE

Aircraft Design Certification GmbH, *Aircraft Design and Certification Engineer* — Oct 2026 – Present
- Aircraft design and certification work within EASA Part 21J and CS-23 frameworks.
- Aircraft and component design using SolidWorks.
- Engineering analysis using OpenFOAM and Femap.

Dynamic Biosensors GmbH, *Working Student in Production* — Oct 2025 – Jul 2026
Munich, Germany
- Supported assembly, verification, and quality assurance of precision biosensor machines in a production environment.
- Collected, documented, and reviewed test data for anomaly tracking, quality documentation, and corrective action support.
- Contributed to process reliability, traceability, and structured technical documentation to support consistent production workflows.
- Worked in a quality-focused environment requiring accuracy, coordination, and disciplined follow-up of technical issues

Microbubbles GmbH, *Working Student Design and Prototyping* — Sep 2024 – Jun 2025
Munich, Germany
- Supported the development of mechanical and aerodynamic assemblies using SolidWorks.
- Fabricated prototype parts using SLA additive manufacturing, milling, and laser cutting.
- Assisted in assembly, experimental testing, and technical documentation.
- Developed Python and VBA tools to improve design iteration and workflow automation.

Turkish Aerospace Industries, *Structural Design Engineer* — Jan 2023 – Jun 2024
Ankara, Türkiye
- Designed aircraft structural components of a vertical stabiliser and wing for a high performance aircraft demonstrator using CATIA V5.
- Produced engineering drawings with GD&T and supported detailed design documentation for manufacturing.
- Performed structural analysis using ANSYS Mechanical to support design development and verification.
- Collaborated with manufacturing and integration activities involving additive manufacturing, milling, laser cutting, and composite-related assembly.

Facility of Advanced Research and Integrated Systems, — Aug 2022 – Dec 2022
Assistant Manager Mechanical — Islamabad, Pakistan
- Led the design of a tilt-wing and tilt-rotor eVTOL UAV prototype, integrating aerodynamic, structural, and manufacturing considerations.
- Conducted preliminary sizing and performance analysis using aircraft design methodologies and MATLAB.
- Developed aircraft assemblies and detailed components in CATIA V5.
- Applied composite design principles in the development of fuselage, wing, and empennage structures.
- Utilised additive manufacturing techniques for rapid prototyping and design validation.

TECHNICAL SKILLS

CAD & Design
CATIA V5, SolidWorks, FEMAP, Openfoam & ANSYS

Analysis & Simulation
MATLAB, Simulink, & Simscape

Manufacturing & Testing
CFRP, Laser Cutting, Milling, & Additive Manufacturing

Documentation — Proficient
SAP, LaTeX, MS Office, & MS PPT

LANGUAGES

Sindhi — Native/Bilingual
Native Language

Urdu — Fluent
C2

English — Native/Bilingual
C2

German — Proficient
B1

CERTIFICATIONS

Drone Pilot License A1/A3 | Aerial Robotics Specialization |
Modern Robotics: Mechanics, Planning, and Control | Additive Manufacturing Specialization

SELECTED PROJECTS

Development and Validation of a Nonlinear Flight Mechanics Model of the AREA Intermeshing Rotor Drone, *Master Thesis at Institute of Rotorcraft and Vertical Flight* — Jul 2026 – Present
Development of a comprehensive nonlinear six-degree-of-freedom flight mechanics model for the AREA unmanned synchropter research platform.

Actuator Fault Detection Strategies for a Quadrotor UAV, — Jun 2025 – Mar 2026
Semester Thesis at Institute of Flight System Dynamics
Investigated model-based actuator fault detection and diagnosis for quadrotors under nominal, faulty, and uncertain conditions utilising observers.

Development of Autonomy Stack for Multirotor in ROS2 — Oct 2025 – Mar 2026
Developed and integrated key autonomy modules for a ROS2–Unity quadrotor simulation stack, including a controller, trajectory planning pipeline, camera workflow, EKF, and mission-level state machine.

Student Team Horyzn, *Flight Testing and Control* — Oct 2024 – Sep 2025
Coordinated safety-critical UAV control system development for a multirotor in the GoAero Safety Challenge, implementing control strategies and MATLAB-based control allocation algorithms to ensure mission-specific stability, performance, and safety.

IMechE UAS Challenge Pakistan, Student Competition — Sep 2022 – Apr 2024
Evaluated UAV concepts and flight demonstrations, coordinated stakeholders, ensured rule compliance and technical documentation, and received the Best Judge Award.

Design and Manufacturing of Fixed Wing CFRP eVTOL UAV — Apr 2021 – Jun 2022
Designed, analysed, manufactured, and validated a fixed-wing eVTOL UAV airframe using SolidWorks, ANSYS FEA, carbon-fibre composites, additive manufacturing, and Pixhawk/ArduPilot flight-control systems.

Design and Manufacturing of Tilt Wing/Tilt Rotor CFRP eVTOL UAV — Sep 2022 – Dec 2022
Designed and validated UAV aerodynamic and structural assemblies using SolidWorks and MATLAB/Simulink, supporting prototyping and airframe manufacturing with carbon-fibre composites and additive manufacturing.

BAJA Society of Automotive Engineers, Student Competition, Arizona USA, — Apr 2019 – Sep 2022
Team Captain
Led a 60-member multidisciplinary engineering team in international vehicle competitions, coordinating chassis, powertrain, suspension, manufacturing, and technical execution while producing GD&T-based engineering drawings and supporting CNC, lathe, laser cutting, and welding processes. Contributed to award-winning results, including 4th place at BAJA SAE Arizona and 2nd overall at the National Piston Cup Challenge.